drones

MDPI

Article

Joint Scheduling and 3D UAV Deployment for Peak Age of Information Minimization in Last-Mile Logistics Networks

Yuting Hao [1] and Jonathan Kua [2,*]

[1] College of Computer and Information Science, Southwest University, Chongqing 400715, China; hyt2021@email.swu.edu.cn
[2] School of Information Technology, Deakin University, Geelong, VIC 3220, Australia
* Correspondence: jonathan.kua@deakin.edu.au

Highlights

What are the main findings?

- A joint offloading association, routing, and 3D UAV deployment framework is proposed to minimize the average Peak Age of Information (PAoI) in smart last-mile logistics networks.
- A BCD-based alternating algorithm is developed. The association and routing subproblem is transformed into an integral single-commodity minimum-cost flow problem, while the UAV deployment subproblems are solved using SCA.
- Under the default setting, the proposed method reduces the average PAoI by approximately 0.6% and the transmission delay by up to 4.2%. The delay reduction exceeds 28% when the benchmark UAV altitude is fixed at 100 m.

What are the implications of the main findings?

- UAV relays can forward selected IoTD data from a heavy-load cell to neighboring light-load BSs, thereby reducing transmission delay and improving information freshness.
- Joint altitude optimization avoids the performance loss caused by an unsuitable fixed altitude, while the observed diminishing returns in the number and service limit of UAVs provide guidance for practical UAV deployment.

Abstract

In smart last-mile logistics networks, real-time status telemetry from ground logistics nodes, such as automated parcel lockers, cold-chain sensors, and unmanned delivery vehicles, is crucial for safe dispatching and precise delivery tracking. However, during peak delivery hours in urban areas, heavy workloads and traffic may lead to severe congestion at base stations (BSs). To address this, multiple unmanned aerial vehicle (UAV) relays are introduced to offload critical logistics monitoring data to neighboring light-load BSs. To quantify the freshness of the received status updates, we adopt the Peak Age of Information (PAoI) metric. We formulate a multi-UAV-assisted multi-cell logistics communication framework that jointly optimizes logistics data offloading association, routing, and 3D UAV deployment to minimize the average PAoI. To solve the resulting mixed-integer non-convex problem, an alternating optimization algorithm based on Block Coordinate Descent (BCD) is developed. Specifically, the offloading association and routing subproblem is reformulated as an integral single-commodity minimum-cost flow problem, while the horizontal UAV positions and altitudes are optimized using Successive Convex Approximation (SCA). Over five random realizations, the proposed method reduces the average PAoI by approximately 0.6% and the transmission delay by up to 4.2% compared with the benchmark methods under the default setting. The delay reduction reaches 6.6% at $K = 60$ and exceeds 28%

Check for updates

Academic Editors: Yoon Seok Chang, Ilkyeong Moon, Chulung Lee and Taesu Cheong

Received: 2 August 2026
Revised: 20 September 2026
Accepted: 30 September 2026
Published: 2 October 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

when the benchmark altitude is fixed at 100 m, demonstrating its effectiveness under high network loads and the importance of UAV altitude optimization.

Keywords: Cellular Internet of Things (C-IoT); multi-UAV systems; Peak Age of Information (PAoI); three-dimensional (3D) deployment; graph theory

1. Introduction

With the rapid growth of e-commerce, healthcare logistics, and urban delivery services, smart last-mile logistics has emerged as an important component of modern intelligent transportation systems. To support automated dispatching, precise parcel tracking, and cold chain status monitoring, ground logistics nodes, such as automated parcel lockers, cold-chain sensors, and unmanned ground vehicles (UGVs), generate a vast amount of time-sensitive telemetry data. Cellular Internet of Things (C-IoT) provides a promising communication solution for these logistics devices because it supports wide coverage, massive connectivity, and low-latency data transmission for mission-critical flows [1–3]. However, logistics telemetry may exhibit uneven traffic demands during peak operating hours. More generally, spatiotemporal variations in cellular traffic demand may create communication hotspots and overload terrestrial cellular networks [4]. Such overload degrades the Quality of Service (QoS) and connectivity of ground logistics nodes [5]. To alleviate traffic congestion at heavy-load BSs, unmanned aerial vehicles (UAVs) can be deployed as aerial relays because of their mobility, rapid deployment, and high probability of line-of-sight (LoS) communication. New network-centric architectures and paradigms have been developed to support UAV autonomy in emerging application domains including smart last-mile logistic networks [6,7].

Recently, UAV-assisted cellular communications in hotspot scenarios have received considerable research attention [8,9]. Traffic-aware UAV deployment strategies were proposed in [4,5] to relieve localized congestion. However, these works focused mainly on direct access without considering flexible relay paths for ground devices. To enhance flexibility, Zeng et al. investigated an Orthogonal Frequency Division Multiple Access (OFDMA) network featuring dual cellular and UAV relay modes to improve throughput [8]. Similarly, Sriharsha et al. proposed UAV-borne Intelligent Reflecting Surfaces (IRSs) coordinated with light-load BSs to offload traffic from heavy-load cells [9]. These studies demonstrate the potential of UAV-assisted traffic offloading, but they do not jointly consider device-level association, end-to-end routing, and 3D multi-UAV deployment. In particular, the selection of an IoTD–UAV–LBS route is coupled with the UAV position through the transmission rates of both hops.

In multi-UAV-assisted smart logistics networks, maintaining high data freshness is crucial. In time-sensitive monitoring and control applications, stale status information reduces the usefulness of the received updates [10]. Consequently, the Age of Information (AoI) metric has been introduced to quantify data freshness. In UAV-assisted networks, Jeganathan et al. optimized average AoI using a full-duplex UAV relay cooperative model [11], while Zhan et al. explored the tradeoff between AoI and operation endurance in multi-cell networks [12]. Several studies have directly considered PAoI in UAV-assisted networks. The works in [13–16] optimized single-UAV trajectories, service sequences, energy allocation, and communication resources for PAoI reduction. These studies establish the importance of UAV mobility and communication resources for PAoI reduction. However, they consider a single mobile UAV and do not address association and routing in a multi-UAV multi-BS network.

Multi-UAV information freshness has also been investigated from different perspectives. Existing studies have jointly considered UAV trajectories, AoI, and device energy consumption [17], graph-based TSP trajectory planning [18], sensing and communication positions with UAV–target matching [19], computation offloading and hovering locations under average peak AoI constraints [20], and secure multi-UAV data collection [21]. These formulations mainly focus on mobile data collection, trajectory planning, or resource allocation and do not optimize destination-BS selection or capacitated IoTD–UAV–LBS routing.

Graph-based optimization has also been applied to UAV-assisted communication networks. Almohamad et al. formulated UAV-assisted multihop backhauling as an integer network-flow problem [22]. However, this model maximizes backhaul flow without considering information freshness, device-level association, or 3D UAV deployment. In the 3D deployment direction, Zeng et al. applied bipartite-graph association and optimized static 3D multi-UAV deployment for video access delay minimization [23]. Although this work considers multi-UAV 3D deployment, its objective is video access delay and its graph model is used for user–UAV association rather than multi-BS routing. Table 1 summarizes the main differences between these closely related studies and this work.

Table 1. Comparison with closely related UAV-assisted studies.

Study	Objective	UAV Setting	Main Optimized Decisions	Role of Graph Optimization	Difference from This Work
[13–15]	PAoI	Single UAV	Trajectory and communication resources	–	No multi-UAV or multi-BS routing
[17]	AoI and energy	Multiple UAVs	Trajectories and device scheduling	–	Mobile collection with one BS
[18]	Average AoI	Multiple UAVs	Flight trajectories	TSP-based trajectory planning	Graph represents UAV tours
[22]	Network flow	Multiple UAVs	UAV-assisted multihop backhauling	Integer network-flow optimization	No PAoI objective or joint 3D deployment
[19]	Average peak AoI	Multiple UAVs	Sensing/communication positions and target matching	UAV–target matching	ISAC rather than multi-BS routing
[20]	Energy with APAoI constraints	Multiple UAVs	Offloading, resources, and hovering locations	–	MEC energy objective
[21]	Average AoI	Multiple UAVs	Secure collection and UAV number	–	No joint 3D deployment and routing
[23]	Video access delay	Multiple UAVs	Association, caching, and static 3D deployment	Bipartite-graph association	No freshness objective or multi-BS routing
This work	Average PAoI	Multiple UAVs	Association, routing, and static 3D deployment	Capacitated minimum-cost-flow routing	Joint device-level multi-BS routing

As shown in Table 1, existing studies cover PAoI optimization, multi-UAV coordination, graph-based planning, and 3D deployment, but address these aspects separately. The existing graph models represent UAV visiting sequences, multihop backhaul flows,

or user–UAV association, rather than complete IoTD service paths with UAV and BS service limits.

To address the aforementioned limitations, this paper presents a joint association, routing, and deployment framework for a multi-UAV-assisted multi-cell network tailored for smart last-mile logistics. Ground logistics devices can use two transmission modes: data can either be uploaded directly to the local heavy-load BS through direct transmission or forwarded through a UAV to a remote light-load BS through UAV relay transmission. We formulate a joint optimization problem to minimize the average PAoI of ground logistics devices by co-optimizing offloading association, routing, and 3D UAV deployment, including horizontal position and flight altitude. The formulated problem is a mixed-integer non-convex optimization problem that is computationally challenging. To solve it efficiently, we develop an iterative optimization algorithm based on Block Coordinate Descent (BCD). The main contributions of this paper are summarized as follows:

- We construct a multi-cell, multi-UAV-relay communication architecture designed for smart last-mile logistics monitoring, enabling selected IoTD data in the heavy-load cell to be forwarded to remote light-load BSs via UAV relays. Each IoTD selects either the direct HBS path or one IoTD–UAV–LBS path. The service limits of the HBS, UAVs, and LBSs are explicitly considered.
- We formulate a joint offloading association, routing, and 3D UAV deployment optimization problem to minimize the average PAoI across all logistics devices, explicitly accounting for altitude-dependent LoS link probabilities and UAV safety spacing constraints.
- We propose an efficient BCD-based alternating optimization algorithm to solve the non-convex problem. Specifically, the discrete offloading association and routing subproblem is mapped onto an integral single-commodity minimum-cost flow problem, while 3D UAV positioning is efficiently updated using SCA. Comprehensive simulation results demonstrate the effectiveness of the proposed algorithm in reducing PAoI compared with benchmark schemes. The simulations also evaluate parameter sensitivity, running time, scalability, and performance under multiple random seeds.

2. System Model and Problem Formulation

2.1. Scenario Description

We consider a multi-cell OFDMA C-IoT network with $M > 1$ rotary-wing UAV relays, which are deployed to assist communication in a heavy-load cell. Specifically, during peak hours, the heavy-load base station (HBS) may be unable to serve all IoTDs in its cell. The LBS set is represented as $\mathcal{L} = \{B_1, \ldots, B_L\}$. To address this issue, M UAV relays are deployed to forward selected IoTD data to the LBSs. The set of UAVs is represented by $\mathcal{M} = \{U_1, \ldots, U_M\}$. The UAVs are homogeneous, and each UAV can serve at most $x_u^{\max}$ IoTDs during the hovering interval. Similarly, the HBS and each LBS can serve at most $x_{bs}^{\max}$ IoTDs. All BSs are deployed at height $H_{bs} > 0$. The horizontal coordinates of the HBS and LBS B_l are denoted by $\mathbf{w}_h = [x_h, y_h]^T \in \mathbb{R}^{2\times 1}$ and $\mathbf{w}_l = [x_l, y_l]^T \in \mathbb{R}^{2\times 1}$, $1 \le l \le L$, respectively, where $[\cdot]^T$ denotes the transpose. During the hovering interval T, UAV U_m is deployed at altitude H_m, and $\mathbf{q}_m = [x_m, y_m]^T \in \mathbb{R}^{2\times 1}$, $1 \le m \le M$, represents its horizontal coordinate. The horizontal position satisfies $\mathbf{q}_m \in \mathcal{Q}$, where $\mathcal{Q} \subseteq \mathbb{R}^2$ denotes the permitted horizontal deployment region. There are K ground IoTDs in the heavy-load cell, and their set is denoted by $\mathcal{K} = \{S_1, \ldots, S_K\}$. We assume that the locations of IoTDs are known to the BSs and UAVs [24,25]. The IoTDs are located at ground level, and the horizontal coordinate of IoTD S_k is denoted by $\mathbf{w}_k = [x_k, y_k]^T \in \mathbb{R}^{2\times 1}$, $1 \le k \le K$.

During the hovering interval T, each IoTD selects one fixed offloading association and routing path for data transmission. For direct transmission, we define the binary variable

$x_{kh} \in \{0,1\}$, where $x_{kh} = 1$ indicates that IoTD S_k transmits directly to the HBS. For relay transmission, we define the binary routing variable $z_{kml} \in \{0,1\}$, where $z_{kml} = 1$ indicates that IoTD S_k transmits via UAV U_m to LBS B_l along the path $S_k \to U_m \to B_l$. Thus, z_{kml} identifies both the serving UAV and the destination LBS for IoTD S_k. Since each IoTD must select exactly one transmission path, the unique-path constraint is given by

$$x_{kh} + \sum_{m=1}^{M} \sum_{l=1}^{L} z_{kml} = 1, \qquad \forall k \in \mathcal{K}. \tag{1}$$

The numbers of IoTDs served by the HBS, each UAV, and each LBS are limited by the following constraints:

$$\sum_{k=1}^{K} x_{kh} \le x_{bs}^{\max}. \tag{2}$$

$$\sum_{k=1}^{K} \sum_{l=1}^{L} z_{kml} \le x_{u}^{\max}, \qquad \forall m \in \mathcal{M}. \tag{3}$$

$$\sum_{k=1}^{K} \sum_{m=1}^{M} z_{kml} \le x_{bs}^{\max}, \qquad \forall l \in \mathcal{L}. \tag{4}$$

2.2. Communication Model

(1) Direct transmission mode: In the direct transmission mode, IoTD S_k transmits its data directly to the HBS over an uplink connection. Since the terrestrial link may be obstructed by buildings, vegetation, and other obstacles, it is modeled as a non-line-of-sight (NLoS) link. The channel power gain from IoTD S_k to the HBS is given by $g_{kh} = \beta_0 \kappa d_{kh}^{-\alpha_N}$, where β_0 denotes the channel power gain at the reference distance $d_0 = 1$ m, $\kappa < 1$ is the additional attenuation factor for the NLoS link, and α_N is the NLoS path-loss exponent. The distance between IoTD S_k and the HBS is $d_{kh} = \sqrt{H_{bs}^2 + \|\mathbf{w}_k - \mathbf{w}_h\|^2}$. Let B denote the bandwidth of each OFDMA subchannel in hertz (Hz). The transmit power of IoTD S_k and the noise power are denoted by p_k and σ^2, respectively. Assuming that orthogonal subchannels are assigned to different links, the achievable uplink rate from IoTD S_k to the HBS is

$$R_{kh} = B \log_2\left(1 + \frac{p_k g_{kh}}{\sigma^2}\right). \tag{5}$$

(2) UAV relay transmission mode: When IoTD S_k selects the UAV relay transmission mode, its data are transmitted to the core network in two stages. First, IoTD S_k transmits the data to UAV U_m. Then, UAV U_m forwards the data to LBS B_l. Following [26], small-scale fading is averaged out, and the large-scale channel is modeled as a probabilistic mixture of line-of-sight (LoS) and NLoS states with distinct path-loss characteristics. For the link between IoTD S_k and UAV U_m, the distance is $d_{km} = \sqrt{H_m^2 + \|\mathbf{w}_k - \mathbf{q}_m\|^2}$, and the corresponding elevation angle is $\theta_{km} = \frac{180}{\pi} \arctan\left(\frac{H_m}{\|\mathbf{w}_k - \mathbf{q}_m\|}\right)$. According to [26], the LoS probability is modeled as $\mathbb{P}_{km}^{L} = B_c + \frac{B_d}{1+\exp[-(B_a + B_b \theta_{km})]}$, where $B_a < 0$, $B_b > 0$, $B_d > 0$, and $B_c = 1 - B_d$ are parameters determined by the propagation environment. The channel power gains under the LoS and NLoS conditions are $g_{km}^{L} = \beta_0 d_{km}^{-\alpha_L}$ and $g_{km}^{N} = \beta_0 \kappa d_{km}^{-\alpha_N}$, respectively. Here, β_0 is the channel power gain at the reference distance, α_L and α_N are the corresponding path-loss exponents, and $\kappa < 1$ is the additional attenuation factor under the NLoS condition. Let $\gamma_{km} = p_k \beta_0 / \sigma^2$. The expected achievable rate is

$$\mathbb{E}[R_{km}] = \mathbb{P}_{km}^{L} B \log_2\left(1 + \frac{\gamma_{km}}{d_{km}^{\alpha_L}}\right) + \left(1 - \mathbb{P}_{km}^{L}\right) B \log_2\left(1 + \frac{\kappa \gamma_{km}}{d_{km}^{\alpha_N}}\right). \tag{6}$$

Since the second term in (6) is nonnegative, removing this term gives the following lower bound:

$$\mathbb{E}[R_{km}] \geq \mathbb{P}_{km}^{L} B \log_2\left(1 + \frac{\gamma_{km}}{d_{km}^{\alpha_L}}\right) \triangleq \widetilde{R}_{km}. \tag{7}$$

As discussed in [27], the LoS rate typically exceeds the NLoS rate because the NLoS link has an additional attenuation factor κ and a larger path-loss exponent α_N. Therefore, $\widetilde{R}_{km}$ is adopted as a conservative lower bound on the expected achievable rate.

The same channel model is used for the link between UAV U_m and LBS B_l. The distance and elevation angle are $d_{ml} = \sqrt{\|H_m - H_{bs}\|^2 + \|\mathbf{q}_m - \mathbf{w}_l\|^2}$ and $\theta_{ml} = \frac{180}{\pi} \arctan\left(\frac{\|H_m - H_{bs}\|}{\|\mathbf{q}_m - \mathbf{w}_l\|}\right)$, respectively. The corresponding LoS probability is $\mathbb{P}_{ml}^{L} = B_c + \frac{B_d}{1+\exp[-(B_a + B_b \theta_{ml})]}$. Accordingly, the rate lower bound for the UAV–LBS link is

$$\widetilde{R}_{ml} = \mathbb{P}_{ml}^{L} B \log_2\left(1 + \frac{\gamma_{ml}}{d_{ml}^{\alpha_L}}\right), \tag{8}$$

where $\gamma_{ml} = p_m^u \beta_0 / \sigma^2$, and p_m^u denotes the transmit power of UAV U_m.

2.3. Peak Age of Information Model

To evaluate the freshness of the received data, we introduce the PAoI metric [28], which is defined as the AoI immediately before a new update is received. We assume that IoTD S_k generates N data packets during the UAV hovering interval T, where $n \in \mathcal{N} = \{1, \ldots, N\}$. The packet generation process of each IoTD is modeled as an independent homogeneous Poisson process with rate λ. Consequently, the interval between two consecutive packet generation times follows an exponential distribution with mean $1/\lambda$. We adopt a queue-free transmission model. Each admitted IoTD transmits its generated packet immediately over its assigned OFDMA resource. Therefore, queueing and processing delays are not considered. The offloading association and routing decisions remain fixed during T.

As shown in Figure 1, t_{kn} and t'_{kn} represent the generation time and reception completion time of the nth packet generated by IoTD S_k, respectively. We define PA_{kn} as the PAoI associated with the nth packet. It is expressed as $PA_{kn} = X_{kn} + Y_{kn}$, where $X_{kn} = t_{kn} - t_{k,n-1}$ is the generation interval between two consecutive packets, and $Y_{kn} = t'_{kn} - t_{kn}$ is the end-to-end transmission delay. All IoTDs use the same packet size for the nth update in each problem instance, i.e., $D_{kn} = D_n$ for all k and n. Based on the two transmission modes and the unique path constraint, PA_{kn} can be written as

$$PA_{kn} = t_{kn} - t_{k,n-1} + x_{kh} \frac{D_n}{R_{kh}} + \sum_{m=1}^{M} \sum_{l=1}^{L} z_{kml} \left(\frac{D_n}{\widetilde{R}_{km}} + \frac{D_n}{\widetilde{R}_{ml}}\right), \qquad n = 2, \ldots, N. \tag{9}$$

The second term in (9) represents the direct transmission delay from IoTD S_k to the HBS. The last term represents the two hop transmission delay through UAV U_m and LBS B_l.

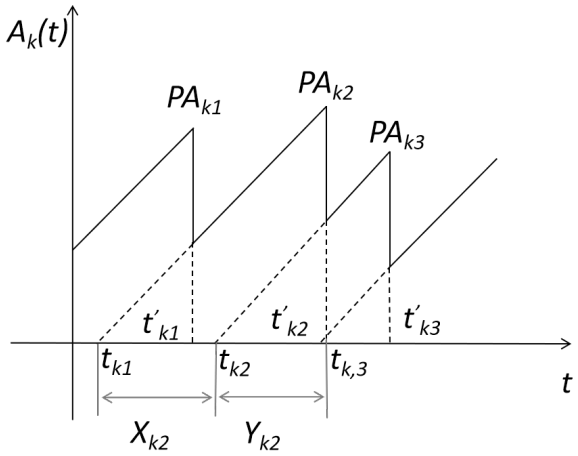


Figure 1. Definition of PAoI for IoTD S_k.

2.4. Problem Formulation

Our objective is to minimize the average PAoI of all packets generated by all IoTDs. We jointly optimize the offloading association and routing decisions, collectively denoted by $\mathbf{X}$, the horizontal UAV positions $\mathbf{Q} = [\mathbf{q}_1, \ldots, \mathbf{q}_M]$, and the UAV altitudes $\mathbf{H} = [H_1, \ldots, H_M]^T$. The optimization problem is formulated as

$$(\mathrm{P1}): \min_{\mathbf{X},\mathbf{Q},\mathbf{H}} \frac{1}{K(N-1)} \sum_{k=1}^{K} \sum_{n=2}^{N} \left[t_{kn} - t_{k,n-1} + x_{kh} \frac{D_n}{R_{kh}} + \sum_{m=1}^{M} \sum_{l=1}^{L} z_{kml} \left(\frac{D_n}{\widetilde{R}_{km}} + \frac{D_n}{\widetilde{R}_{ml}} \right) \right] \tag{10a}$$

$$\text{s.t.} \quad x_{kh} \in \{0,1\}, \qquad \forall k \in \mathcal{K}, \tag{10b}$$

$$z_{kml} \in \{0,1\}, \qquad \forall k \in \mathcal{K},\ m \in \mathcal{M},\ l \in \mathcal{L}, \tag{10c}$$

$$x_{kh} + \sum_{m=1}^{M} \sum_{l=1}^{L} z_{kml} = 1, \qquad \forall k \in \mathcal{K}, \tag{10d}$$

$$\sum_{k=1}^{K} x_{kh} \le x_{bs}^{\max}, \tag{10e}$$

$$\sum_{k=1}^{K} \sum_{l=1}^{L} z_{kml} \le x_{u}^{\max}, \qquad \forall m \in \mathcal{M}, \tag{10f}$$

$$\sum_{k=1}^{K} \sum_{m=1}^{M} z_{kml} \le x_{bs}^{\max}, \qquad \forall l \in \mathcal{L}, \tag{10g}$$

$$\mathbf{q}_m \in \mathcal{Q}, \qquad \forall m \in \mathcal{M}, \tag{10h}$$

$$\|\mathbf{q}_i - \mathbf{q}_j\|_2 \ge d_{\min}, \qquad 1 \le i < j \le M, \tag{10i}$$

$$H_{\min} \le H_m \le H_{\max}, \qquad \forall m \in \mathcal{M}. \tag{10j}$$

Constraints (10b) and (10c) define the binary association and routing variables. Constraint (10d) ensures that each IoTD selects exactly one transmission path. Constraints (10e)–(10g) limit the maximum number of IoTDs served by the HBS, each UAV, and each LBS, respectively. Constraint (10h) limits each UAV to the permitted horizontal deployment region. Constraint (10i) ensures the minimum horizontal separation $d_{\min}$ between any two UAVs. Constraint (10j) limits the UAV altitudes between $H_{\min}$ and $H_{\max}$.

Problem (10) is a mixed-integer nonconvex problem because of the binary variables, nonconvex rate functions, and minimum UAV separation constraint. To solve this problem, we develop a BCD-based alternating optimization algorithm. Problem (10) is decomposed into three subproblems: association and routing optimization, horizontal UAV deployment optimization, and UAV altitude optimization. The association and routing subproblem is transformed into a minimum-cost flow problem, while the two deployment subproblems are solved using SCA. The three subproblems are solved sequentially until convergence.

3. The Proposed Algorithm

3.1. Offloading Association and Routing Optimization

To simplify problem (10), we first optimize the IoTD offloading association and routing decisions for a given 3D UAV deployment. Since the packet generation interval $t_{kn} - t_{k,n-1}$ is independent of these decisions, it does not affect the optimal solution and is omitted from the objective function of this subproblem. Therefore, problem (10) is reduced to

$$\begin{aligned} (\text{P2}): \quad \min_{\mathbf{X}} \quad & \frac{1}{K(N-1)} \sum_{k=1}^{K} \sum_{n=2}^{N} \left[x_{kh} \frac{D_n}{R_{kh}} \right. \\ & \left. + \sum_{m=1}^{M} \sum_{l=1}^{L} z_{kml} \left(\frac{D_n}{\widetilde{R}_{km}} + \frac{D_n}{\widetilde{R}_{ml}} \right) \right] \\ \text{s.t.} \quad & (10b)\text{–}(10g). \end{aligned} \tag{11}$$

Problem (11) can be formulated as a single-commodity minimum-cost flow problem. Since each IoTD uses one fixed transmission path during the hovering interval, its path selection is represented by one unit of flow. Thus, if f units of flow pass through a UAV or a BS, that node serves f IoTDs. The service limits $x_u^{\max}$ and $x_{bs}^{\max}$ can therefore be represented by flow capacity constraints.

Based on this correspondence, we construct an augmented directed graph $\widetilde{G} = (\widetilde{V}, \widetilde{E})$. Its node set contains one super source node v_s, K IoTD nodes, two nodes v_m^{in} and v_m^{out} for each UAV, one HBS node, L LBS nodes, and one super sink node v_t. Therefore, $|\widetilde{V}| = K + 2M + L + 3$. Let $\bar{D} = \frac{1}{N-1} \sum_{n=2}^{N} D_n$ denote the average packet size. The transmission costs of the IoTD–HBS, IoTD–UAV, and UAV–LBS links are defined as $c_{kh} = \bar{D}/R_{kh}$, $c_{km} = \bar{D}/\widetilde{R}_{km}$, and $c_{ml} = \bar{D}/\widetilde{R}_{ml}$, respectively. The super source is connected to each IoTD node by an edge with unit capacity and zero cost. Each IoTD node is connected to the HBS node by an edge with unit capacity and cost c_{kh}, and to each UAV input node by an edge with unit capacity and cost c_{km}. For each UAV, an internal edge from v_m^{in} to v_m^{out} has capacity $x_u^{\max}$ and zero cost. Each UAV output node is connected to every LBS node by an edge with capacity $x_{bs}^{\max}$ and cost c_{ml}. Finally, the HBS node and each LBS node are connected to the super sink by edges with capacity $x_{bs}^{\max}$ and zero cost. The super source supplies K units of flow, and the super sink has a demand of K units. Let f_{ij}, u_{ij}, and c_{ij} denote the flow, capacity, and cost of edge $(i,j) \in \widetilde{E}$, respectively. Problem (11) is then equivalent to

$$\begin{aligned} (\text{P3}): \quad \min_{\{f_{ij}\}} \quad & \frac{1}{K} \sum_{(i,j) \in \widetilde{E}} c_{ij} f_{ij} \\ \text{s.t.} \quad & 0 \le f_{ij} \le u_{ij}, \qquad \forall (i,j) \in \widetilde{E}, \\ & \sum_{j:(i,j) \in \widetilde{E}} f_{ij} - \sum_{j:(j,i) \in \widetilde{E}} f_{ji} = b_i, \qquad \forall i \in \widetilde{V}, \end{aligned} \tag{12}$$

where $b_{v_s} = K$, $b_{v_t} = -K$, and $b_i = 0$ for all other nodes.

Since all supplies, demands, and edge capacities are integer-valued, the integrality property of minimum-cost flow problems guarantees that problem (12) admits an integer

optimal solution. The super source supplies K units of flow through K super-source–IoTD edges, each with unit capacity. Together with the flow conservation constraints, this ensures that each IoTD sends one unit of flow along exactly one transmission path. A direct path through the HBS corresponds to $x_{kh} = 1$, whereas a relay path through UAV U_m and LBS B_l corresponds to $z_{kml} = 1$. Therefore, the integer flow solution gives a feasible binary association and routing solution.

3.2. UAV Horizontal Deployment Optimization

Given the association and routing decisions $\mathbf{X}$ and UAV altitudes $\mathbf{H}$, we optimize the horizontal UAV positions $\mathbf{Q}$. Let $\bar{D} = \frac{1}{N-1}\sum_{n=2}^{N} D_n$ denote the average packet size. The packet generation term and the direct transmission delay are independent of $\mathbf{Q}$. We combine these two fixed terms as

$$C_{\text{fix}} = \frac{1}{K(N-1)} \sum_{k=1}^{K} \sum_{n=2}^{N} (t_{kn} - t_{k,n-1}) + \frac{\bar{D}}{K} \sum_{k=1}^{K} \frac{x_{kh}}{R_{kh}}. \tag{13}$$

The horizontal deployment subproblem is formulated as

$$\begin{aligned} (\text{P4}): \quad \min_{\mathbf{Q}} \quad & C_{\text{fix}} + \frac{\bar{D}}{K} \sum_{k=1}^{K} \sum_{m=1}^{M} \sum_{l=1}^{L} z_{kml} \left(\frac{1}{\widetilde{R}_{km}} + \frac{1}{\widetilde{R}_{ml}} \right) \\ \text{s.t.} \quad & \mathbf{q}_m \in \mathcal{Q}, \qquad \forall m \in \mathcal{M}, \\ & \|\mathbf{q}_i - \mathbf{q}_j\|_2^2 \geq d_{\min}^2, \qquad 1 \leq i < j \leq M. \end{aligned} \tag{14}$$

Problem (14) is nonconvex because the rate functions depend nonlinearly on $\mathbf{Q}$ and the minimum UAV separation constraint is nonconvex. We apply SCA to construct a convex approximation [29]. For the links selected by the current routing solution, we introduce an epigraph variable η, positive reciprocal-rate variables y_{km} and y_{ml}, and auxiliary elevation-angle variables $\theta_{km}, \theta_{ml} \in [0, 90]$. The variable η upper-bounds the average PAoI in the approximated problem, while y_{km} and y_{ml} upper-bound the reciprocals of the corresponding rate lower bounds.

At iteration r, define $\mathbf{v}_{km} = [\theta_{km}, \mathbf{q}_m^T]^T$, and let $\mathbf{v}_{km}^r = [\theta_{km}^r, (\mathbf{q}_m^r)^T]^T$ denote the current local point. Let L_{km}^{2D} be an upper bound on the spectral norm of the corresponding rate Hessian over the feasible deployment region. According to the smoothness inequality [30], the IoTD–UAV rate satisfies

$$\begin{aligned} \widetilde{R}_{km}(\mathbf{v}_{km}) \geq \; & \widetilde{R}_{km}(\mathbf{v}_{km}^r) + \nabla \widetilde{R}_{km}(\mathbf{v}_{km}^r)^T (\mathbf{v}_{km} - \mathbf{v}_{km}^r) \\ & - \frac{L_{km}^{\text{2D}}}{2} \|\mathbf{v}_{km} - \mathbf{v}_{km}^r\|_2^2 \triangleq \widetilde{R}_{km}^{\text{lb},r}. \end{aligned} \tag{15}$$

Similarly, $\widetilde{R}_{ml}^{\text{lb},r}$ is constructed using $\mathbf{v}_{ml} = [\theta_{ml}, \mathbf{q}_m^T]^T$ and the corresponding curvature constant L_{ml}^{2D}.

The auxiliary elevation angles must be related to the horizontal UAV positions. Let $r_{km}^r = \|\mathbf{w}_k - \mathbf{q}_m^r\|_2$. Since the elevation angle is convex with respect to the horizontal link distance, its first-order approximation gives the following lower bound:

$$g_{km}^{\text{lb},r}(\mathbf{q}_m) = \theta_{km}^r - \frac{180 H_m}{\pi[(r_{km}^r)^2 + H_m^2]} (\|\mathbf{w}_k - \mathbf{q}_m\|_2 - r_{km}^r). \tag{16}$$

The lower bound $g_{ml}^{\text{lb},r}$ is constructed in the same way for the UAV–LBS link using its horizontal distance and vertical distance $H_m - H_{bs}$.

Since the squared distance between two UAVs is convex, its first-order approximation gives the following affine lower bound:

$$\begin{aligned}\Lambda_{ij}^{\mathrm{lb},r} &= 2(\mathbf{q}_i^r - \mathbf{q}_j^r)^T(\mathbf{q}_i - \mathbf{q}_j) - \|\mathbf{q}_i^r - \mathbf{q}_j^r\|_2^2 \\ &\le \|\mathbf{q}_i - \mathbf{q}_j\|_2^2.\end{aligned} \tag{17}$$

Using these lower bounds, problem (14) is approximated at iteration r as

$$\begin{aligned}(\mathrm{P5}): \quad \min_{\mathbf{Q},\boldsymbol{\theta},\mathbf{y},\eta} \quad & \eta \\ \text{s.t.} \quad & C_{\mathrm{fix}} + \frac{\bar{D}}{K}\sum_{k=1}^{K}\sum_{m=1}^{M}\sum_{l=1}^{L} z_{kml}(y_{km}+y_{ml}) \le \eta, \\ & \frac{1}{y_{km}} \le \widetilde{R}_{km}^{\mathrm{lb},r}, \quad \forall (k,m) \text{ with } \sum_{l=1}^{L} z_{kml} > 0, \\ & \frac{1}{y_{ml}} \le \widetilde{R}_{ml}^{\mathrm{lb},r}, \quad \forall (m,l) \text{ with } \sum_{k=1}^{K} z_{kml} > 0, \\ & \theta_{km} \le g_{km}^{\mathrm{lb},r}, \quad \forall (k,m) \text{ with } \sum_{l=1}^{L} z_{kml} > 0, \\ & \theta_{ml} \le g_{ml}^{\mathrm{lb},r}, \quad \forall (m,l) \text{ with } \sum_{k=1}^{K} z_{kml} > 0, \\ & \Lambda_{ij}^{\mathrm{lb},r} \ge d_{\min}^2, \qquad 1 \le i < j \le M, \\ & \mathbf{q}_m \in \mathcal{Q}, \qquad \forall m \in \mathcal{M}.\end{aligned} \tag{18}$$

It can be observed that the rate and elevation-angle lower bounds are concave, while $\Lambda_{ij}^{\mathrm{lb},r}$ is affine. Therefore, all constraints in problem (18) are convex, and the resulting horizontal deployment problem can be solved using CVX.

3.3. UAV Altitude Optimization

Given the association and routing decisions $\mathbf{X}$ and horizontal UAV positions $\mathbf{Q}$, we optimize the UAV altitudes $\mathbf{H}$. The packet generation term and the direct transmission delay remain constant and are represented by C_{fix} in (13). The altitude optimization subproblem is formulated as

$$\begin{aligned}(\mathrm{P6}): \quad \min_{\mathbf{H}} \quad & C_{\mathrm{fix}} + \frac{\bar{D}}{K}\sum_{k=1}^{K}\sum_{m=1}^{M}\sum_{l=1}^{L} z_{kml}\left(\frac{1}{\widetilde{R}_{km}} + \frac{1}{\widetilde{R}_{ml}}\right) \\ \text{s.t.} \quad & H_{\min} \le H_m \le H_{\max}, \qquad \forall m \in \mathcal{M}.\end{aligned} \tag{19}$$

The minimum UAV separation constraint is not included because it depends only on the fixed horizontal positions.

Problem (19) is nonconvex because the link rates depend nonlinearly on the UAV altitudes. For the selected links, we introduce the same epigraph, reciprocal-rate, and auxiliary elevation-angle variables as in the horizontal deployment subproblem.

At iteration r, define $\mathbf{u}_{km} = [\theta_{km}, H_m]^T$ and $\mathbf{u}_{ml} = [\theta_{ml}, H_m]^T$. Let L_{km}^{H} and L_{ml}^{H} be upper bounds on the spectral norms of the corresponding rate Hessians over the feasible altitude region. The smoothness inequality gives

$$\begin{aligned}\widetilde{R}_{km}^{\mathrm{lb},r} &= \widetilde{R}_{km}(\mathbf{u}_{km}^r) + \nabla\widetilde{R}_{km}(\mathbf{u}_{km}^r)^T(\mathbf{u}_{km} - \mathbf{u}_{km}^r) - \frac{L_{km}^{\mathrm{H}}}{2}\|\mathbf{u}_{km} - \mathbf{u}_{km}^r\|_2^2, \\ \widetilde{R}_{ml}^{\mathrm{lb},r} &= \widetilde{R}_{ml}(\mathbf{u}_{ml}^r) + \nabla\widetilde{R}_{ml}(\mathbf{u}_{ml}^r)^T(\mathbf{u}_{ml} - \mathbf{u}_{ml}^r) - \frac{L_{ml}^{\mathrm{H}}}{2}\|\mathbf{u}_{ml} - \mathbf{u}_{ml}^r\|_2^2.\end{aligned} \tag{20}$$

For fixed horizontal positions, let $h_{km}(H_m)$ and $h_{ml}(H_m)$ denote the physical elevation angles of the IoTD–UAV and UAV–LBS links, respectively. We assume $H_{\min} > H_{bs}$. The absolute values of their second derivatives are bounded by

$$\ell_{km} = \frac{180}{\pi}\frac{9}{8\sqrt{3}H_{\min}^2}, \qquad \ell_{ml} = \frac{180}{\pi}\frac{9}{8\sqrt{3}(H_{\min}-H_{bs})^2}. \tag{21}$$

According to the smoothness inequality, the corresponding concave quadratic lower bounds are

$$\begin{aligned} h_{km}^{\mathrm{lb},r}(H_m) &= h_{km}(H_m^r) + h'_{km}(H_m^r)(H_m - H_m^r) - \frac{\ell_{km}}{2}(H_m - H_m^r)^2, \\ h_{ml}^{\mathrm{lb},r}(H_m) &= h_{ml}(H_m^r) + h'_{ml}(H_m^r)(H_m - H_m^r) - \frac{\ell_{ml}}{2}(H_m - H_m^r)^2. \end{aligned} \tag{22}$$

Using these lower bounds, problem (19) is approximated at iteration r as

$$\begin{aligned} (\mathrm{P7}): \quad \min_{\mathbf{H},\boldsymbol{\theta},\mathbf{y},\eta} \quad & \eta \\ \text{s.t.} \quad & C_{\mathrm{fix}} + \frac{\bar{D}}{K}\sum_{k=1}^{K}\sum_{m=1}^{M}\sum_{l=1}^{L} z_{kml}(y_{km}+y_{ml}) \le \eta, \\ & \frac{1}{y_{km}} \le \widetilde{R}_{km}^{\mathrm{lb},r}, \quad \forall (k,m) \text{ with } \sum_{l=1}^{L} z_{kml} > 0, \\ & \frac{1}{y_{ml}} \le \widetilde{R}_{ml}^{\mathrm{lb},r}, \quad \forall (m,l) \text{ with } \sum_{k=1}^{K} z_{kml} > 0, \\ & \theta_{km} \le h_{km}^{\mathrm{lb},r}(H_m), \quad \forall (k,m) \text{ with } \sum_{l=1}^{L} z_{kml} > 0, \\ & \theta_{ml} \le h_{ml}^{\mathrm{lb},r}(H_m), \quad \forall (m,l) \text{ with } \sum_{k=1}^{K} z_{kml} > 0, \\ & H_{\min} \le H_m \le H_{\max}, \qquad \forall m \in \mathcal{M}. \end{aligned} \tag{23}$$

Similarly, the quadratic rate and elevation-angle lower bounds are concave functions of the altitude variables. Hence, problem (23) is convex and is solved by CVX at each altitude SCA iteration.

3.4. Overall Alternating Algorithm, Convergence, and Computational Cost

The overall procedure is summarized in Algorithm 1. The association and routing decisions, horizontal UAV positions, and UAV altitudes are alternately updated in the order $\mathbf{X} \to \mathbf{Q} \to \mathbf{H}$. Let $J^r = \overline{PA}(\mathbf{X}^r, \mathbf{Q}^r, \mathbf{H}^r)$ denote the average PAoI after the rth outer iteration. For fixed $\mathbf{Q}^r$ and $\mathbf{H}^r$, solving problem (12) does not increase the objective value. Moreover, each candidate obtained from problem (18) or (23) is accepted only if it does not increase the original PAoI. Therefore,

$$\begin{aligned} J^r &\ge \overline{PA}(\mathbf{X}^{r+1}, \mathbf{Q}^r, \mathbf{H}^r) \\ &\ge \overline{PA}(\mathbf{X}^{r+1}, \mathbf{Q}^{r+1}, \mathbf{H}^r) \\ &\ge \overline{PA}(\mathbf{X}^{r+1}, \mathbf{Q}^{r+1}, \mathbf{H}^{r+1}) = J^{r+1}. \end{aligned} \tag{24}$$

Since the average PAoI is nonnegative, the sequence $\{J^r\}$ is non-increasing and bounded below. Hence, the objective value converges. Since problem (12) is solved using `intlinprog`, while problems (18) and (23) are solved using CVX, the practical computational cost depends on the problem dimensions and the numbers of inner and outer iterations. Therefore, it is evaluated through the running time and scalability experiments.

Algorithm 1 Overall BCD algorithm for association, routing, and UAV 3D deployment

1: Initialize feasible horizontal positions $\mathbf{Q}^0$ and altitudes $\mathbf{H}^0$.
2: Given $\mathbf{Q}^0$ and $\mathbf{H}^0$, solve problem (12) to obtain $\mathbf{X}^0$, and calculate $J^0 = \overline{PA}(\mathbf{X}^0, \mathbf{Q}^0, \mathbf{H}^0)$.
3: Set $r = 0$, convergence tolerance ϵ_{out}, and maximum iteration number $R_{\max}$.
4: **repeat**
5: Given $\mathbf{Q}^r$ and $\mathbf{H}^r$, solve problem (12) to obtain $\mathbf{X}^{r+1}$.
6: Given $\mathbf{X}^{r+1}$ and $\mathbf{H}^r$, solve problem (18) to obtain a candidate horizontal deployment. Set it as $\mathbf{Q}^{r+1}$ if it does not increase the original PAoI; otherwise, set $\mathbf{Q}^{r+1} = \mathbf{Q}^r$.
7: Given $\mathbf{X}^{r+1}$ and $\mathbf{Q}^{r+1}$, solve problem (23) to obtain a candidate altitude deployment. Set it as $\mathbf{H}^{r+1}$ if it does not increase the original PAoI; otherwise, set $\mathbf{H}^{r+1} = \mathbf{H}^r$.
8: Calculate $J^{r+1} = \overline{PA}(\mathbf{X}^{r+1}, \mathbf{Q}^{r+1}, \mathbf{H}^{r+1})$.
9: Set $r \leftarrow r + 1$.
10: **until** $r \geq R_{\max}$ or $\dfrac{|J^r - J^{r-1}|}{\max\{1, |J^{r-1}|\}} \leq \epsilon_{\text{out}}$
11: **Output:** $\mathbf{X}^r$, $\mathbf{Q}^r$, and $\mathbf{H}^r$.

4. Experimental Results

4.1. Simulation Setup

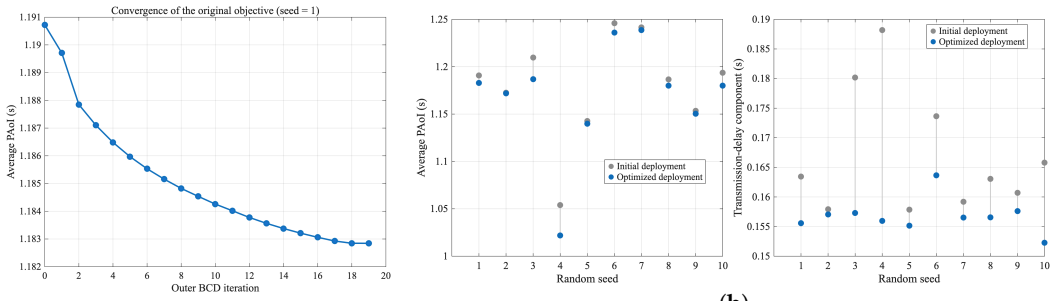
We consider a multi-cell C-IoT network in a 1 km × 1 km square area. The horizontal coordinate of the HBS is $[250, 250]^T$ m, while those of the three LBSs are $[650, 650]^T$ m, $[200, 780]^T$ m, and $[720, 200]^T$ m. All BSs are deployed at a height of $H_{bs} = 25$ m. Unless otherwise stated, we consider $M = 2$ UAVs and $K = 50$ IoTDs. The IoTDs are uniformly distributed within the intersection of the considered square region and the HBS coverage disk with a radius of 300 m. The horizontal UAV deployment region is $\mathcal{Q} = [0, 1000] \times [0, 1000]\,\text{m}^2$. The packet arrival rate is $\lambda = 1$ packet/s, the UAV hovering interval is $T = 10$ s, and each IoTD generates $N = 5$ updates. The first update is generated at time zero, and the remaining interarrival times follow an exponential distribution. A realization is retained only if the Nth update of every IoTD is generated within T. The default packet size is $D_n = 1$ Mbit. The HBS and each LBS can serve at most $x_{bs}^{\max} = 40$ IoTDs, while each UAV can serve at most $x_u^{\max} = 10$ IoTDs. The UAV altitude satisfies 100 m $\leq H_m \leq$ 400 m, and its initial value is 250 m. The default minimum horizontal separation between two UAVs is $d_{\min} = 5$ m. The LoS probability parameters are $B_a = -0.4568$, $B_b = 0.0470$, $B_c = -0.63$, and $B_d = 1.63$ [23]. The remaining communication parameters are $B = 1$ MHz, $p_k = p_m^u = 0.1$ W, $\kappa = 0.2$, $\alpha_L = 2$, $\alpha_N = 2.2$, $\sigma^2 = -110$ dBm, and $\beta_0 = -50$ dB. Unless otherwise stated, each experiment is repeated using five random seeds. The figures report the sample mean, and the error bars represent one standard deviation. The packet arrival-rate experiment instead reports the 95% confidence interval. All simulations are performed using MATLAB R2025b and CVX 2.2 on a MacBook Air equipped with an Apple M5 processor and 16 GB of memory.

4.2. Algorithm Convergence and Computational Time

The stopping tolerances of the outer BCD iteration, horizontal SCA iteration, and altitude SCA iteration are set to $\epsilon_{\text{out}} = 10^{-5}$, $\epsilon_{\text{2D}} = 10^{-6}$, and $\epsilon_{\text{H}} = 10^{-6}$, respectively. For the convergence experiment, the maximum numbers of these iterations are all set to 100.

(1) Convergence behavior: Figure 2a shows that the average PAoI decreases and becomes stable after 19 outer iterations, illustrating the numerical convergence behavior of Algorithm 1. To further evaluate the convergence behavior, the algorithm is tested using ten random realizations. Figure 2b compares the objective values before and after optimization for each realization. The required number of outer iterations ranges from 3 to 38, with an average of 14.1 iterations. Moreover, all intermediate solutions are feasible, and the

maximum recorded constraint violation is zero within numerical precision. These results demonstrate the numerical convergence behavior and feasibility of the proposed algorithm.



(**a**) (**b**)

Figure 2. Convergence results: (**a**) PAoI versus iteration; (**b**) initial and optimized performance over ten random realizations.

(2) Computational time and scalability: Figure 3 shows the running time of the proposed algorithm as the number of IoTDs increases. Each point is averaged over five random realizations, and the error bars represent one standard deviation. Since each outer iteration requires solving the association and routing subproblem and two SCA subproblems, the maximum number of outer iterations is set to 30 to provide the same computational budget for all problem instances. As K increases from 45 to 60, the average running time increases from 619.8 s to 4494.2 s, while the average number of outer iterations increases from 15.4 to 30. The observed running time grows faster than linearly over the considered range. This increase is caused by both the enlarged optimization subproblems and the greater number of outer iterations.

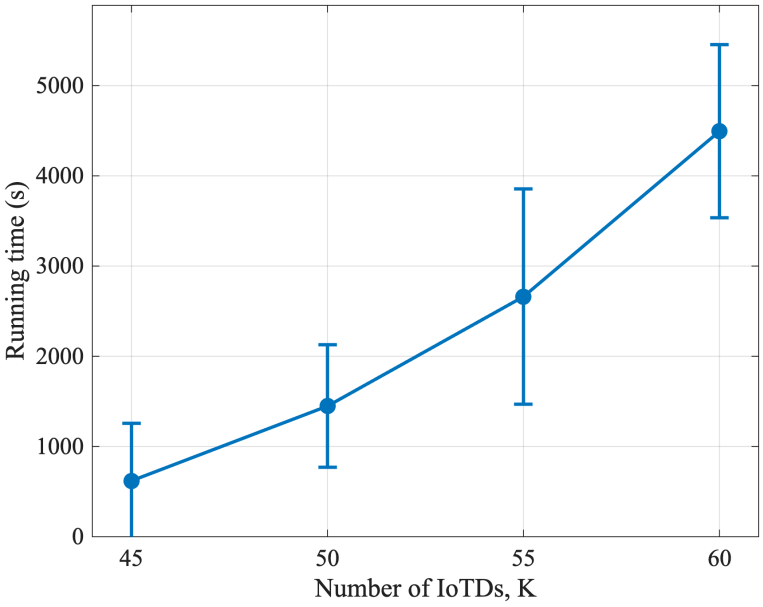


Figure 3. Running time of the proposed algorithm versus the number of IoTDs.

(3) Sensitivity to feasible initialization: Under the same network realization, five feasible random UAV deployments generated using different initialization seeds are tested. As shown in Figure 4, the final average PAoI ranges from 1.1786 s to 1.1843 s, with a coefficient of variation of approximately 0.22%. This indicates that the final objective value is relatively insensitive to the tested initial deployments. The running time ranges from 2.1 s to 2315.4 s because different initial deployments require different numbers of iterations. Thus, the

initial deployment has a limited effect on the final PAoI but can significantly affect the convergence speed.

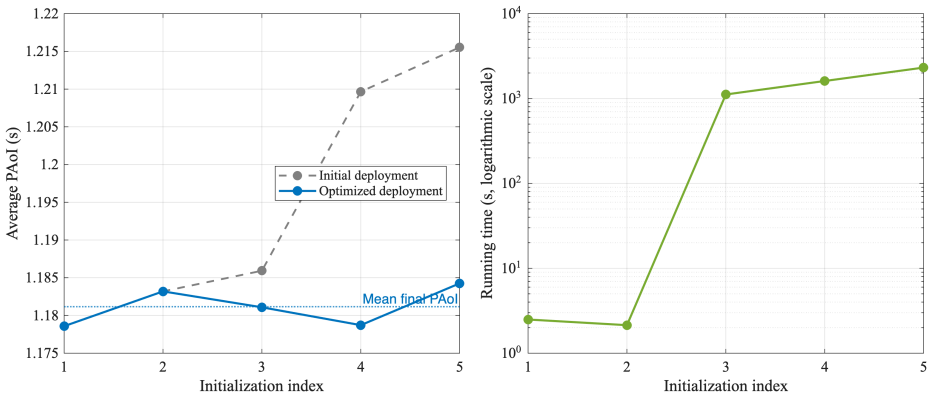
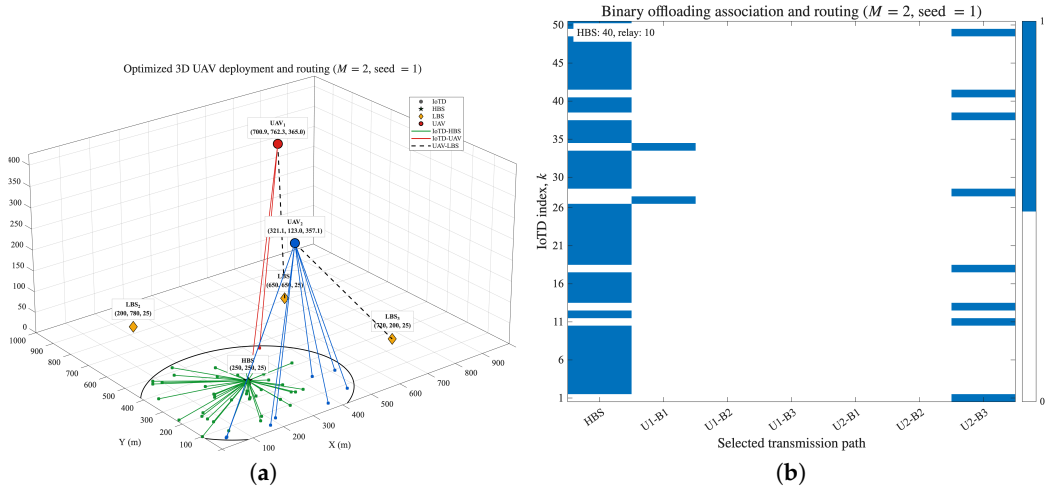


Figure 4. Final average PAoI and running time under different feasible random initial deployments.

4.3. Optimized Deployment and Routing Decisions

Figure 5 presents the optimized deployment and routing decisions for a representative realization with $M = 2$, $K = 50$, and random seed 1. In Figure 5a, the green lines represent direct IoTD–HBS links, the colored solid lines represent IoTD–UAV links, and the dashed lines represent UAV–LBS links. Figure 5b shows the corresponding binary routing variables, where blue and white cells denote one and zero, respectively. Among the 50 IoTDs, 40 transmit directly to the HBS, two use the path through U_1 and B_1, and eight use the path through U_2 and B_3. Each row contains exactly one nonzero entry, which confirms the unique-path constraint.



(**a**) (**b**)

Figure 5. Optimized deployment and routing decisions for $M = 2$: (**a**) 3D UAV deployment and selected transmission paths; (**b**) binary association and routing decisions.

4.4. Performance Comparison

We compare the proposed method with the K-means and 2D deployment methods. The K-means method determines $\mathbf{Q}$ from the IoTD cluster centers and optimizes $\mathbf{X}$ at a fixed altitude. The 2D deployment method jointly optimizes $\mathbf{X}$ and $\mathbf{Q}$ while fixing $\mathbf{H}$. In contrast, the proposed method jointly optimizes $\mathbf{X}$, $\mathbf{Q}$, and $\mathbf{H}$.

Table 2 presents the results for $M = 2$ and $K = 50$. Each result is averaged over five random realizations and reported as the sample mean and standard deviation. As can be observed from Table 2, the performance difference is particularly evident at low UAV altitudes. At 100 m, the proposed method reduces the transmission delay by 28.73% and 28.25% compared with the K-means and 2D deployment methods, respectively. When the

fixed altitude increases from 100 m to 250 m, the transmission delays of the two benchmark methods decrease by 25.78% and 25.08%. Therefore, the performance gap narrows considerably at 250 m and becomes even smaller at 400 m. This trend results from the joint effects of the UAV altitude on the LoS probability and propagation distance. Increasing the altitude increases the elevation angle and hence the LoS probability, but it also increases the link distance. Within the considered altitude range, the rate improvement provided by the increased LoS probability dominates the additional distance loss. At higher altitudes, the marginal increase in LoS probability becomes smaller, and different horizontal deployments produce smaller relative differences in the 3D link distances. Consequently, the results of the three methods gradually become closer. Nevertheless, the proposed method achieves the lowest mean PAoI and transmission delay by jointly optimizing $\mathbf{X}$, $\mathbf{Q}$, and $\mathbf{H}$.

Table 2. Performance comparison under optimized and fixed UAV altitudes.

Method	UAV Altitude (m)	Average PAoI (s)	Transmission Delay (s)
Proposed 3D	Optimized	1.1406 ± 0.0690	0.1562 ± 0.0009
K-means	100	1.2036 ± 0.0693	0.2192 ± 0.0121
	175	1.1591 ± 0.0676	0.1747 ± 0.0055
	250	1.1471 ± 0.0670	0.1627 ± 0.0039
	325	1.1430 ± 0.0668	0.1586 ± 0.0036
	400	1.1418 ± 0.0667	0.1573 ± 0.0035
2D deployment	100	1.2021 ± 0.0959	0.2177 ± 0.0370
	175	1.1591 ± 0.0799	0.1747 ± 0.0172
	250	1.1475 ± 0.0745	0.1631 ± 0.0099
	325	1.1478 ± 0.0624	0.1634 ± 0.0101
	400	1.1454 ± 0.0632	0.1609 ± 0.0080

4.5. Impact of System Parameters

We further evaluate the impact of the main system parameters on the average PAoI and transmission delay component. Unless otherwise specified, only the parameter shown on the horizontal axis is varied, while the other parameters retain their default values. Figure 6 shows the impact of the packet arrival rate. As λ increases, the mean interarrival time $1/\lambda$ decreases, thereby reducing the average PAoI. In contrast, the transmission-delay component of each method remains unchanged because the arrival process is independent of the association, routing, and deployment variables under the adopted queue-free model.

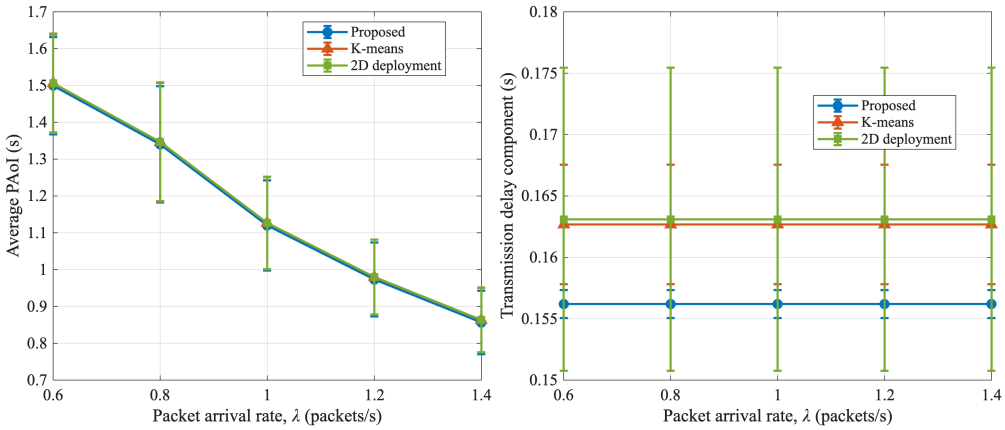


Figure 6. Impact of the packet arrival rate on the average PAoI and transmission delay component.

An inverse relationship between the system bandwidth and both the average PAoI and transmission delay is observed in Figure 7, with larger values of B corresponding to smaller values of both metrics. This trend is attributed to the higher achievable link rates provided by a larger bandwidth. Under the adopted fixed-noise-power model, all transmission delays scale inversely with B.

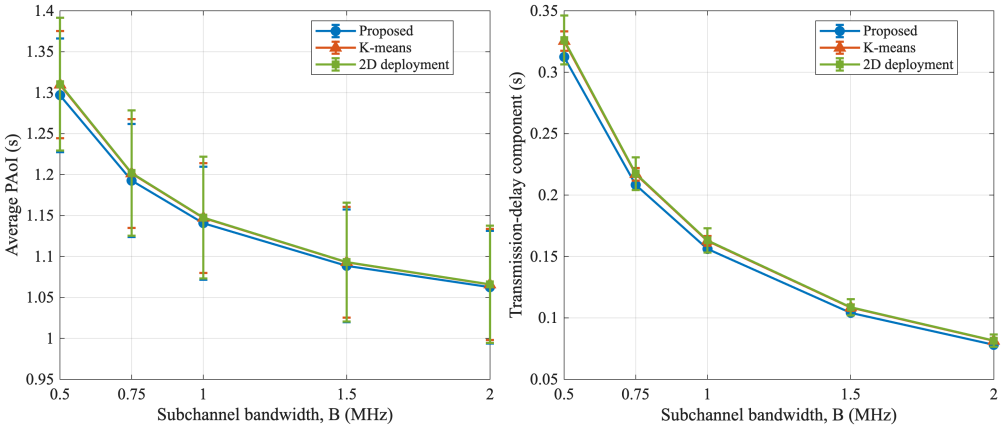


Figure 7. Impact of the system bandwidth on the average PAoI and transmission delay component.

We further investigate how the minimum UAV separation affects the system performance. The parameter $d_{\min}$ determines the required horizontal distance between the two UAVs. We set $K = 55$ to ensure that both UAVs are activated by the routing solution. As shown in Figure 8, the average PAoI and transmission delay remain nearly unchanged when $d_{\min}$ increases from 5 to 100 m. This result indicates that the optimized distance between the UAVs already satisfies the considered separation requirements. Therefore, the safety-distance constraint is inactive, and increasing $d_{\min}$ within this range has little effect on the system performance.

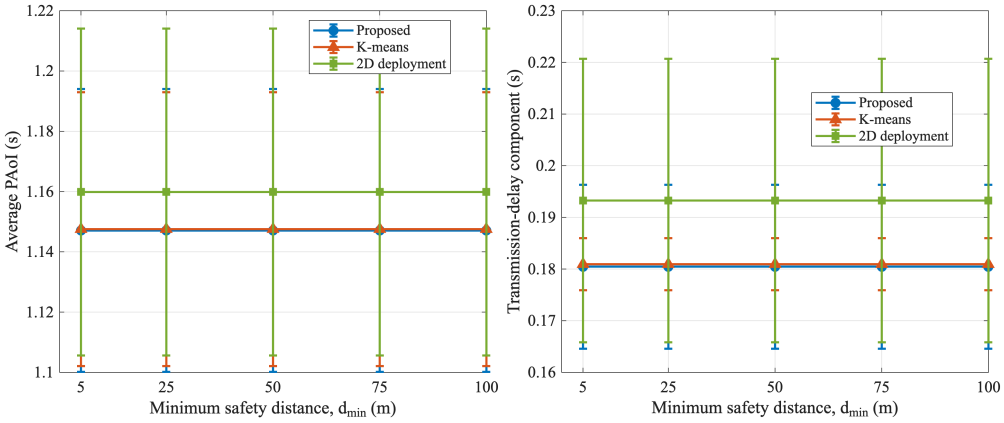


Figure 8. Impact of the minimum UAV separation on the average PAoI and transmission delay component for $K = 55$.

Figure 9 illustrates the influence of the number of IoTDs on the performance of the three methods. The results indicate that both the average PAoI and transmission delay increase with K. Specifically, the transmission delay of the proposed method increases from 0.1461 s at $K = 45$ to 0.1824 s at $K = 60$. This is because more IoTDs compete for the limited service capacities of the UAVs and BSs, reducing the flexibility of association and routing decisions. Nevertheless, the proposed method consistently achieves the lowest mean values. Its transmission-delay reductions over the K-means and 2D deployment methods increase from 1.9% and 2.1% at $K = 45$ to 6.6% and 5.4% at $K = 60$, respectively.

These results demonstrate the advantage of the proposed joint optimization under higher network loads.

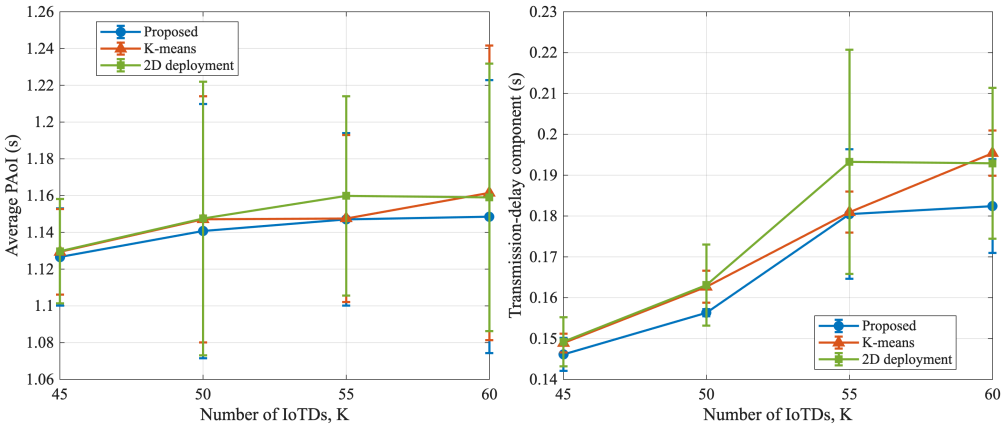


Figure 9. Impact of the number of IoTDs on the average PAoI and transmission delay component.

The influence of the number of UAVs is shown in Figure 10. It is observed that the average PAoI and transmission delay decrease as M increases. This is because more UAVs provide additional relay choices for the IoTDs and increase the available relay service capacity. However, the performance improvement gradually diminishes when the available UAV resources become sufficient for the considered IoTDs.

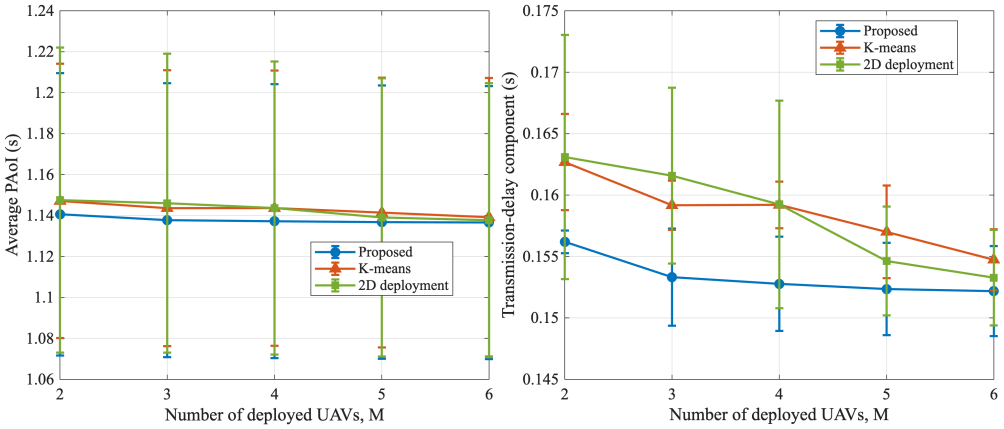


Figure 10. Impact of the number of UAVs on the average PAoI and transmission delay component.

Figure 11 examines the effect of the UAV service limit. Increasing $x_u^{\max}$ allows more IoTDs to use relay paths, thereby reducing the average PAoI and transmission delay. However, both metrics remain almost unchanged when $x_u^{\max} \geq 10$. This occurs because the available UAV capacity is already sufficient for the IoTDs selected for relay transmission. Therefore, further increasing the service limit provides no additional performance gain under the default setting.

A positive relationship between the packet size and the two performance metrics is observed in Figure 12. According to the transmission-time term D_n/R, increasing D_n enlarges the delay of each selected link and consequently raises the average PAoI. By jointly adjusting the routing decisions and UAV deployment, the proposed method limits this increase more effectively than the benchmark methods, leading to a larger absolute performance gap for larger packets.

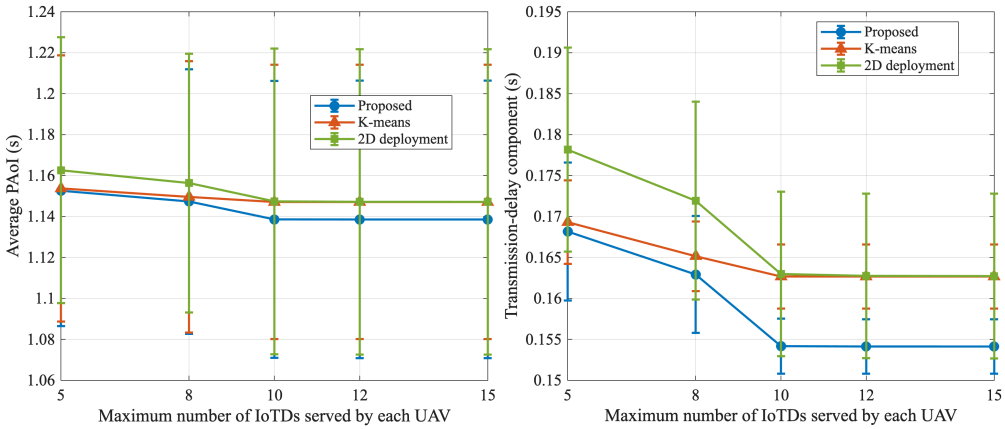


Figure 11. Impact of the UAV service limit on the average PAoI and transmission delay component.

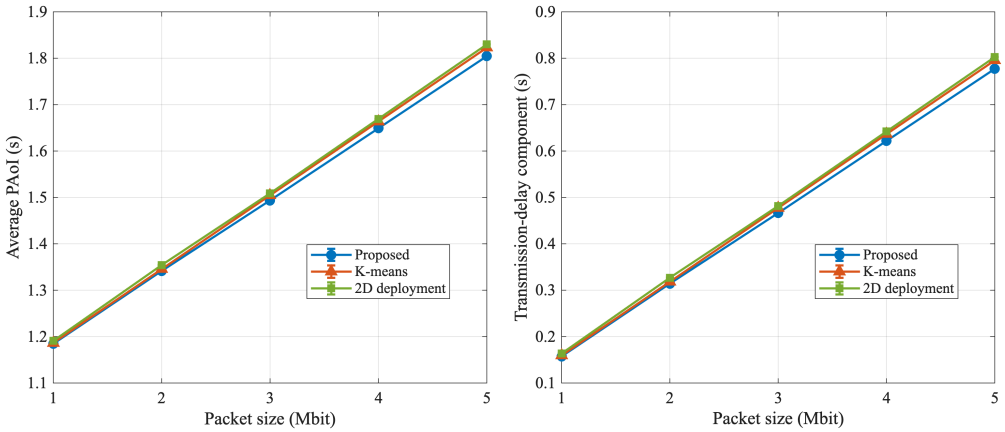


Figure 12. Impact of the packet size on the average PAoI and transmission delay component.

5. Conclusions and Future Work

In this paper, we investigated the joint offloading association, routing, and 3D UAV deployment problem for PAoI minimization in a multi-cell C-IoT network. To solve the resulting mixed-integer non-convex problem, we developed a BCD-based alternating algorithm. The association and routing subproblem was transformed into a single-commodity minimum-cost flow problem, while the horizontal UAV positions and altitudes were optimized using SCA. Simulation results show that the proposed method achieves lower mean PAoI and transmission delay than the K-means and 2D deployment methods. At the default benchmark altitude of 250 m, the transmission-delay reductions are 3.98% and 4.23%, respectively. When the number of IoTDs increases to $K = 60$, the reductions reach 6.6% and 5.4%, indicating a greater advantage under a higher network load. When the benchmark altitude is fixed at 100 m, the corresponding reductions increase to 28.73% and 28.25%. These results demonstrate the benefits of jointly optimizing association, routing, and 3D UAV deployment, particularly under high network loads or an unfavorable fixed altitude.

The current model assumes fixed association and routing decisions during the hovering interval, queue-free packet transmission, and known device locations. Future work can extend this study in the following directions:

- **Queue-Aware PAoI Optimization:** Packet queueing, overlapping updates, processing delay, and adaptive packet generation can be incorporated into the PAoI model.
- **Dynamic Trajectory and Energy Optimization:** UAV trajectories, propulsion energy, battery constraints, and charging decisions can be jointly optimized under time-varying traffic and channel conditions.

- **Online and Robust Control:** Online or distributed algorithms can be developed for imperfect device locations, uncertain channel information, and dynamic network states.

Author Contributions: Conceptualization, Y.H. and J.K.; methodology, Y.H.; software, Y.H.; validation, Y.H. and J.K.; formal analysis, Y.H.; investigation, Y.H.; resources, J.K.; data curation, Y.H.; writing—original draft preparation, Y.H.; writing—review and editing, J.K.; visualization, Y.H.; supervision, J.K.; project administration, J.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The simulation data and MATLAB code are available from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Moges, T.H.; Lakew, D.S.; Nguyen, N.P.; Dao, N.N.; Cho, S. Cellular Internet of Things: Use cases, technologies, and future work. *Internet Things* **2023**, *24*, 100910. [CrossRef]
2. Wang, S.; Kua, J.; Jin, J.; Wong, Y.W.; Jayaraman, P.P.; Pang, Z. Assisting mission-critical traffic flows with active queue management in Industrial Internet of Things. *J. Ind. Inf. Integr.* **2026**, 101170. [CrossRef]
3. Ladegourdie, M.; Kua, J. Performance analysis of OPC UA for industrial interoperability towards industry 4.0. *IoT* **2022**, *3*, 507–525. [CrossRef]
4. Wang, Y.; Yan, M.; Feng, G.; Qin, S.; Wei, F. Autonomous On-Demand Deployment for UAV Assisted Wireless Networks. *IEEE Trans. Wirel. Commun.* **2023**, *22*, 9488–9501. [CrossRef]
5. Mahapatra, B.; Verma, A.; Gupta, D.; Sharma, P.K.; Turuk, A.K. Traffic-Aware UAV Placement Strategies for Load Balancing in 5G Cellular Hotspots. In *2021 Advanced Communication Technologies and Signal Processing (ACTS)*; IEEE: New York, NY, USA, 2021; pp. 1–6. [CrossRef]
6. Jin, J.; Yu, K.; Kua, J.; Zhang, N.; Pang, Z.; Han, Q.-L. Cloud-Fog Automation: Vision, Enabling Technologies, and Future Research Directions. *IEEE Trans. Ind. Inform.* **2024**, *20*, 1039–1054. [CrossRef]
7. Jin, J.; Pang, Z.; Kua, J.; Zhu, Q.; Johansson, K.H.; Marchenko, N.; Cavalcanti, D. Cloud-fog automation: The new paradigm towards autonomous industrial cyber-physical systems. *IEEE J. Sel. Areas Commun.* **2025**. [CrossRef]
8. Zeng, S.; Zhang, H.; Di, B.; Song, L. Trajectory optimization and resource allocation for OFDMA UAV relay networks. *IEEE Trans. Wirel. Commun.* **2021**, *20*, 6634–6647. [CrossRef]
9. Sriharsha, C.; Murthy, C.S.R. A Novel Cellular User Offloading via UAV-Borne IRS. *IEEE Wirel. Commun. Lett.* **2023**, *12*, 1736–1740. [CrossRef]
10. Yates, R.D.; Sun, Y.; Brown, D.R.; Kaul, S.K.; Modiano, E.; Ulukus, S. Age of Information: An Introduction and Survey. *IEEE J. Sel. Areas Commun.* **2021**, *39*, 1183–1210. [CrossRef]
11. Jeganathan, A.; Dhayabaran, B.; Perera, T.D.P.; Jayakody, D.N.K.; Muthuchidambaranathan, P. Harvest-on-sky: An AoI-driven UAV-assisted wireless communication system. *IEEE Internet Things Mag.* **2022**, *5*, 142–146. [CrossRef]
12. Zhan, C.; Hu, H.; Wang, J.; Liu, Z.; Mao, S. Tradeoff Between Age of Information and Operation Time for UAV Sensing Over Multi-Cell Cellular Networks. *IEEE Trans. Mob. Comput.* **2024**, *23*, 2976–2991. [CrossRef]
13. Abd-Elmagid, M.A.; Dhillon, H.S. Average peak age-of-information minimization in UAV-assisted IoT networks. *IEEE Trans. Veh. Technol.* **2019**, *68*, 2003–2008. [CrossRef]
14. Cao, A.; Shen, C.; Zong, J.; Chang, T.H. Peak age-of-information minimization of UAV-aided relay transmission. In *2020 IEEE International Conference on Communications Workshops (ICC Workshops)*; IEEE: New York, NY, USA, 2020; pp. 1–6. [CrossRef]
15. Jiang, W.; Shen, C.; Ai, B. UAV-assisted sensing and communication design for average peak age-of-information minimization. In *2022 IEEE International Conference on Communications Workshops (ICC Workshops)*; IEEE: New York, NY, USA, 2022; pp. 1005–1010. [CrossRef]
16. Jia, D.; Kua, J.; Liu, X. A lightweight LSTM model for flight trajectory prediction in autonomous UAVs. *Future Internet* **2025**, *18*, 4. [CrossRef]
17. Eldeeb, E.; de Souza Sant'Ana, J.M.; Echevarría Pérez, D.; Shehab, M.; Mahmood, N.H.; Alves, H. Multi-UAV Path Learning for Age and Power Optimization in IoT with UAV Battery Recharge. *IEEE Trans. Veh. Technol.* **2023**, *72*, 5356–5360. [CrossRef]

18. Rahimi, O.; Shafieinejad, A. Minimizing Age of Information in Multi-UAV-Assisted IoT Networks: A Graph Theoretical Approach. *Wirel. Netw.* **2024**, *30*, 533–555. [CrossRef]
19. Yu, B.; Yang, W.; Wang, Q.; Zhang, R.; Cai, Y. Age of Information Analysis and Optimization in Unmanned Aerial Vehicles-Assisted Integrated Sensing and Communication Systems. *J. Electron. Inf. Technol.* **2024**, *46*, 1996–2003. [CrossRef]
20. Shen, S.; Yang, H.; Yang, K.; Wang, K.; Zhang, G. AoI-Aware Joint Resource Allocation in Multi-UAV Aided Multi-Access Edge Computing Systems. *IEEE Trans. Netw. Sci. Eng.* **2024**, *11*, 2596–2609. [CrossRef]
21. Tekin, N. Age of Information Minimization for Secure Data Collection in Multi-UAV-Assisted IoT Applications. *Internet Things* **2025**, *33*, 101672. [CrossRef]
22. Almohamad, A.; Hasna, M.O.; Khattab, T.; Haouari, M. On Network Flow Maximization via Multihop Backhauling and UAVs: An Integer Programming Approach. In *2019 IEEE 89th Vehicular Technology Conference (VTC2019-Spring)*; IEEE: New York, NY, USA, 2019; pp. 1–6. [CrossRef]
23. Zeng, B.; Zhan, C.; Xu, C.; Liao, J. Caching and 3D Deployment Strategy for Scalable Videos in Cache-Enabled Multi-UAV Networks. *IEEE Trans. Veh. Technol.* **2023**, *72*, 14875–14888. [CrossRef]
24. Malik, Y.; Abdul Khaliq, K.; Abdulrazak, B.; Tariq, U. Mobile Node Localization in Cellular Networks. *Int. J. Wirel. Mob. Netw.* **2011**, *3*, 91–100. [CrossRef]
25. Tahar, K.N.; Kamarudin, S.S. UAV Onboard GPS in Positioning Determination. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2016**, *XLI-B1*, 1037–1042. [CrossRef]
26. You, C.; Zhang, R. Hybrid offline-online design for UAV-enabled data harvesting in probabilistic LoS channels. *IEEE Trans. Wirel. Commun.* **2020**, *19*, 3753–3768. [CrossRef]
27. Xu, C.; Zhan, C.; Yang, H.; Xiao, L. Pareto-Optimal Aerial-Ground Energy Minimization for Aerial 3D Mobile Edge Computing Networks. *IEEE Trans. Veh. Technol.* **2024**, *73*, 7218–7232. [CrossRef]
28. Sun, Y.; Kadota, I.; Talak, R.; Modiano, E. *Age of Information: A New Metric for Information Freshness*; Springer Nature: Berlin/Heidelberg, Germany, 2022.
29. Scutari, G.; Facchinei, F.; Lampariello, L. Parallel and Distributed Methods for Constrained Nonconvex Optimization—Part I: Theory. *IEEE Trans. Signal Process.* **2017**, *65*, 1929–1944. [CrossRef]
30. Nesterov, Y. *Introductory Lectures on Convex Optimization: A Basic Course*; Applied Optimization; Kluwer Academic Publishers: Boston, MA, USA, 2004; Volume 87. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.